



MEGN 300: Spring 2026
Design Challenge One: Sprint-3

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Design Challenge 1: The Scale - Part 1 Analysis

Abstract:

The goal of this project is to successfully develop a way to convert voltage readings to mass in grams, utilizing a custom designed scale and a LabView VI. A set range of required mass readings have been assigned to each team ensuring variation in design approaches. Several scale designs and sensor recommendations were provided, but it is up to each team to decide how they will integrate clear design intent, calibration techniques and LabView into their scales.

Introduction and design:

Our scale uses a lever arm, spring, and nichrome wire to measure a mass between 0 and 150 grams. This design applies Newton's second law and Hooke's law to relate the displacement of the spring at one end of the scale to the mass at the other. In order to measure the displacement of the spring, a metal plate is positioned at the spring end of the scale to make constant contact with a segment of nichrome wire. Using the known resistivity of the wire, the known resistance of a resistor in series with the wire, and the voltage measured across the nichrome by the LabVIEW DAQ, the voltage divider equation can be used to determine the displacement of the spring (shown in the circuit diagram in Figure 1). Mechanically, the design will feature a parallel bar mechanism (depicted below in Figure 2) to keep the mass and the contact plate steady while it rotates.

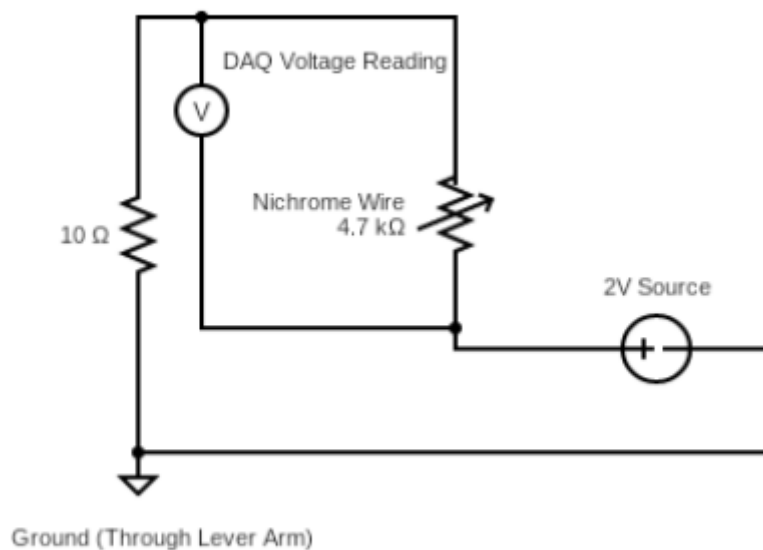


Figure 1: Wiring Diagram - Nichrome Wire

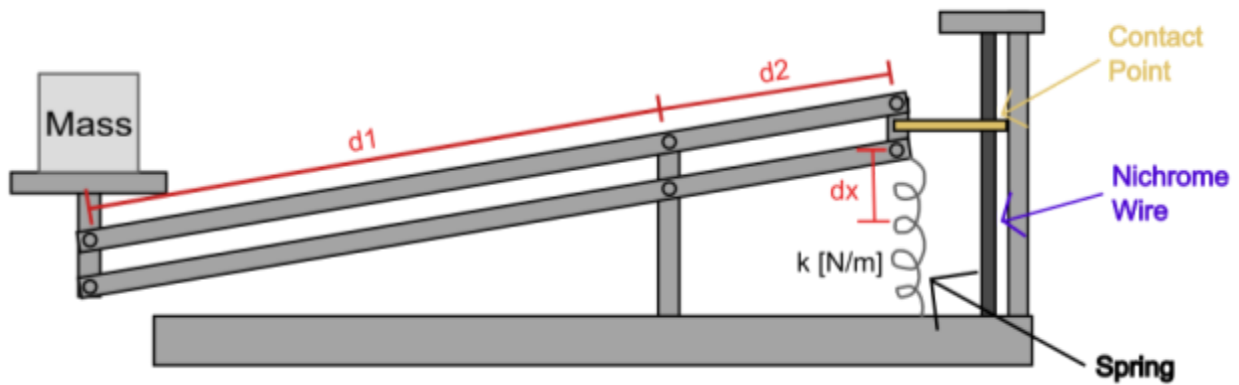


Figure 2: Theoretical Scale Design

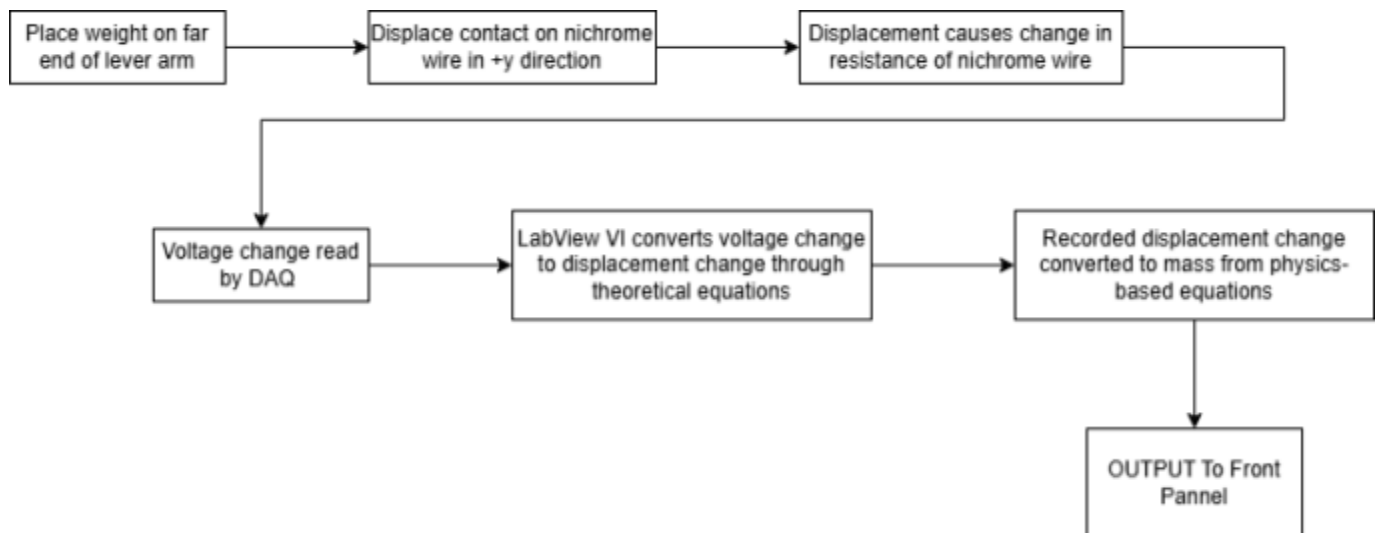


Figure 3: Flow Diagram

Analysis and design considerations:

Several parameters can be changed to optimize the readings produced by the scale. Those parameters include the resistivity of the nichrome wire (ρ), the distances between the mass/spring and the pivot point (d_1 & d_2), the resistance of the resistor in series with the nichrome wire (R_{NC}), the spring constant (k), and the input voltage (V_0).

The mass in question can be related to scale response first through the displacement of the spring at the through Newton's second law. An FBD of the scale bar is provided below in Figure 4 for visualization.

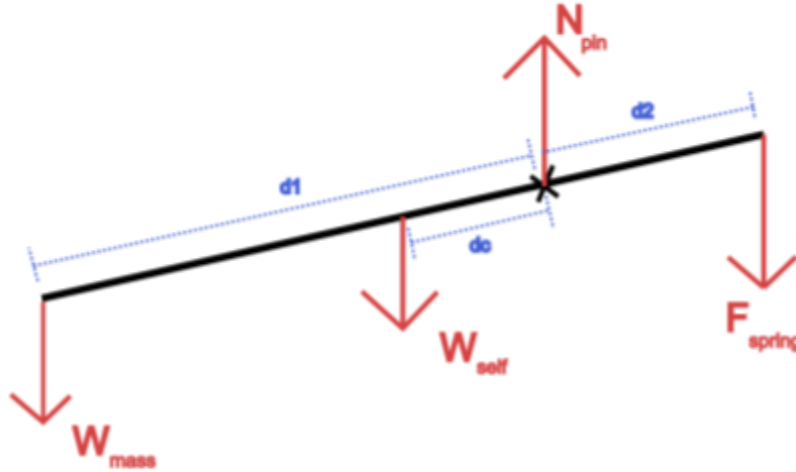


Figure 4: FBD of scale given an applied weight

$$\Sigma M = 0 = W_{mass} * d_1 + W_{self} - F_s * d_2 + N_{pin}$$

$$\Sigma M = 0 = (m * g * d_1) + (m_{self} * g * d_c) - (k * \Delta x * d_2)$$

Isolating mass reveals its relationship with spring displacement:

$$m = \frac{k * \Delta x * d_2 - m_{self} * g * d_c}{g * d_1}$$

The spring displacement can then be related to the voltage across a segment of the nichrome wire (the response voltage) through a voltage divider equation and the resistivity of the nichrome ($R_{NC} = \rho * \Delta x$). The voltage divider equation with the substituted resistivity in the numerator is the following:

$$V_{NC} = \frac{R_{NC}}{R_{NC} + R_1} * V_0 = \frac{\rho * \Delta x}{\rho * \Delta x + R_1} * V_0$$

Isolating displacement yields the following equation:

$$\Delta x = \frac{R_1 * V_{NC}}{\rho * (V_0 - V_{NC})}$$

Plugging this equation into the equation for mass produces the final equation that relates the change in response voltage (V_{NC}) to the mass:

$$m = \frac{k \cdot d_2 \cdot \left(\frac{R_1 \cdot V_{NC}}{\rho \cdot (V_0 - V_{NC})} \right) - m_{self} \cdot g \cdot d_c}{g \cdot d_1}$$

Using this equation, we generated a theoretical calibration curve using several reasonable values for our parameters tabulated below in Table 1. These parameters can be easily updated if our design changes to produce an accurate theoretical calibration curve.

Table 1: Parameter Values

Parameter	Value
g [gravity]	9.81 [m/s^2]
p [resistivity]	0.26 [Ω /m]
d1 [weight distance]	0.5 [m]
d2 [spring distance]	0.25 [m]
dc [CoM distance]	0.125 [m]
R1 [known resistor]	10 [Ω]
k [spring constant]	0.5 [N/m]
V0 [input voltage]	2 [V]
m _{self} [mass of scale bars]	0.1 [kg]

The calibration curve shown below in Figure 5 demonstrates a clear proportional relationship between mass and the sensor response operating within the given range for mass for the design requirements (0-150 grams). The voltage response is notably large enough to be precisely measured by the DAQ.

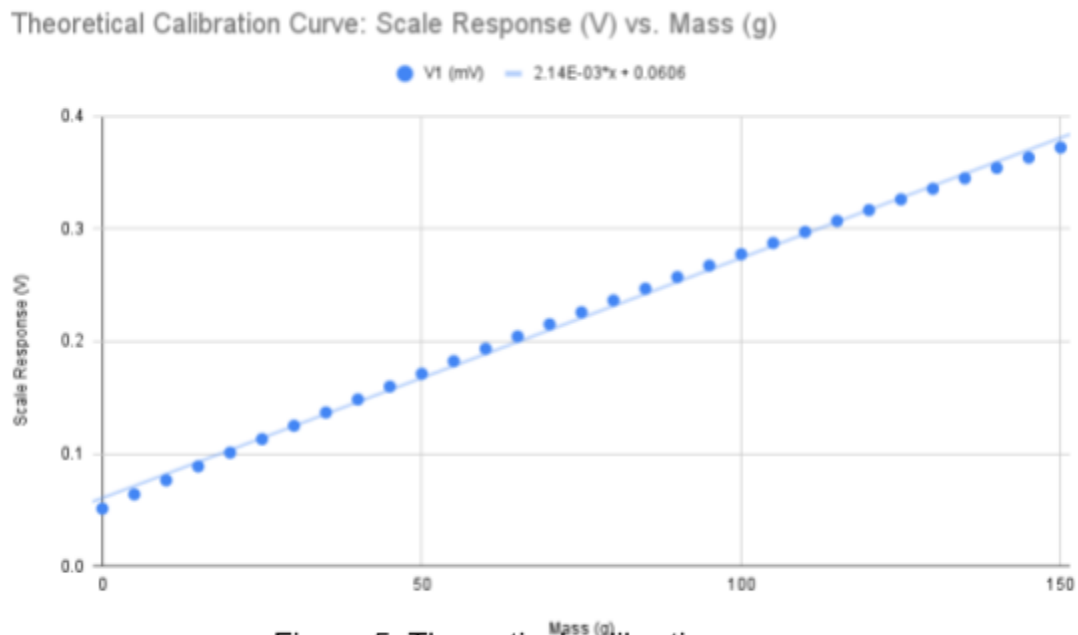


Figure 5: Theoretical calibration curve

Summary of Operation:

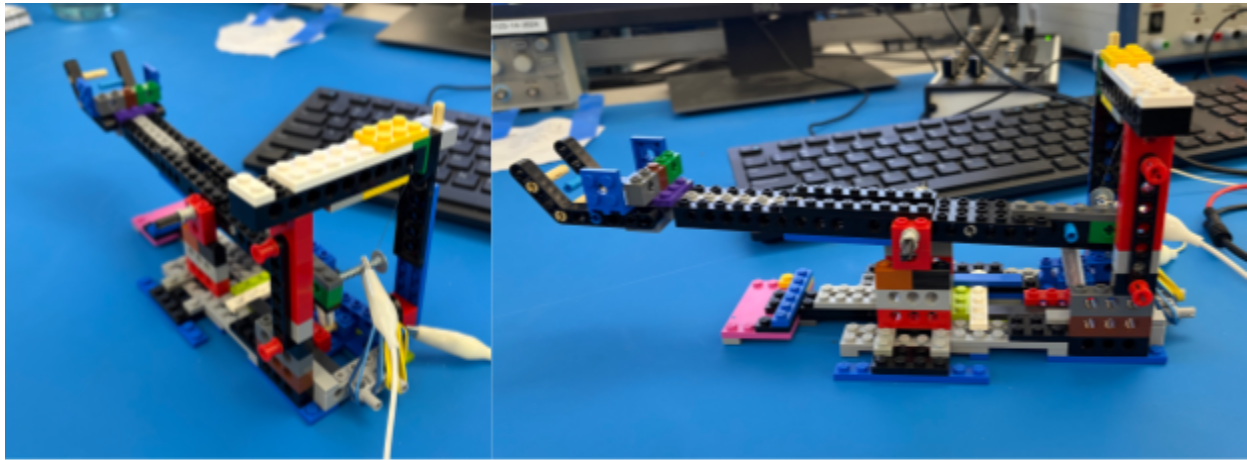


Figure 6: Final Scale Design and Configuration

As pictured above in Figure 6, the final scale design was constructed from miscellaneous pieces of lego, a small length of Nichrome 80 wire, a small bolt to act as the contact point, and several rubber bands. We initially struggled with maintaining consistent contact between the bolt and the nichrome; the nichrome would often lose tension, or the bolt would drag the length of nichrome as it moved. To resolve this issue, we wrapped several rubber bands around the base of the nichrome wire to create constant tension.

Operating the scale is relatively straightforward. After ensuring that the electronics are properly configured—including the placement of a 10 ohm shunt resistor in series with the nichrome contact and two alligator clips measuring the voltage across the contact—simply tare the scale using the tare toggle on the front panel (pictured below in Figure 7). This will set the “Converted Mass (g)” indicator to zero, at which point the user can begin placing mass in the basket on the left while observing an accurate reading of the mass in that same indicator.

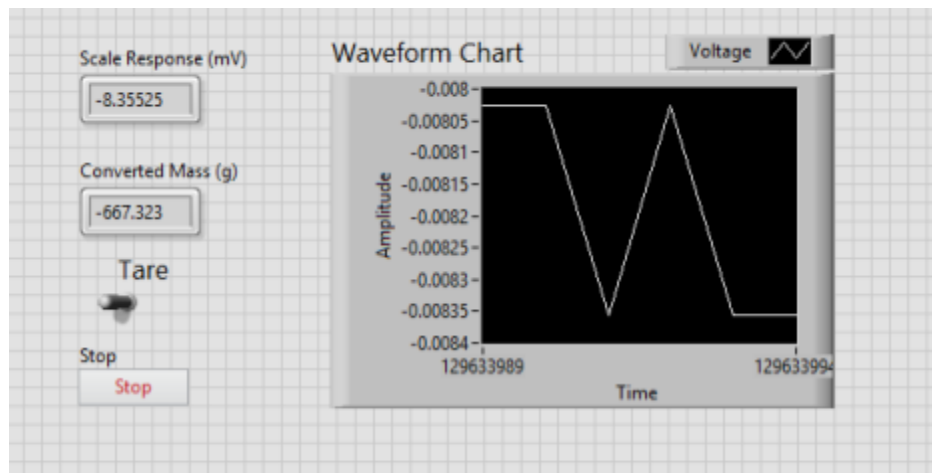


Figure 7: Scale LabVIEW Front Panel

This tare feature is crucial to the design, as it helps to eliminate bias error caused by some mechanical issues with the scale (e.g. the nichrome wire losing tension over time or the bolt

slipping down the nichrome to a different location). Figure 8 below shows the primary VI used to accurately measure the mass placed on the end of the scale. It should be noted that a separate Sub VI is used to convert the scale response voltage to mass using the equation generated by our experimental calibration curve (visible in the appendix in Figure a-2).

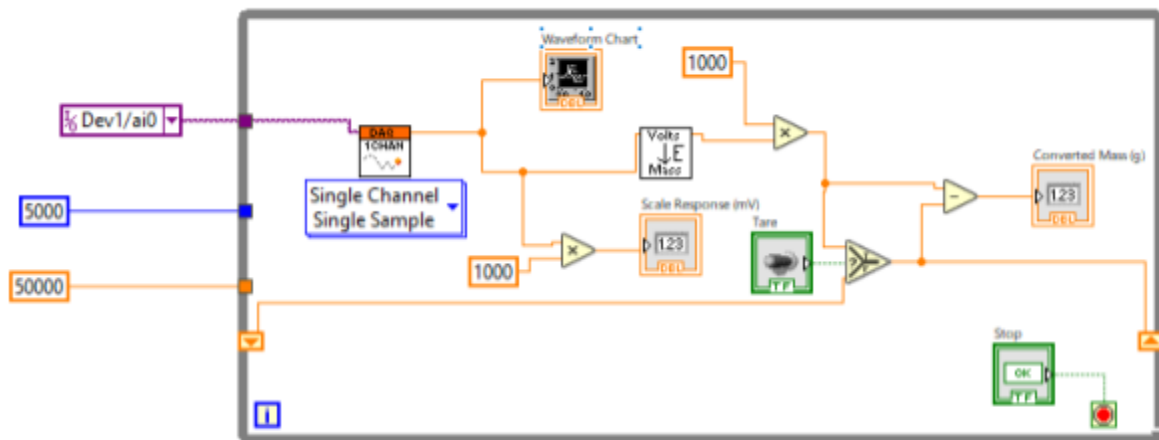


Figure 8: Scale VI Block Diagram with Tare Feature

This VI initializes the port on the DAQ with leads across the scale's contact. This voltage data is passed through the conversion Sub VI (appendix Figure a-2) to yield a reasonable mass value. If the tare control is activated, this mass is passed through all subsequent loops with a shift register to be subtracted from every following mass reading. Every time the tare control is activated, this value is updated. After subtraction, the final mass value is passed to the "Converted Mass (g)" indicator, which updates on the front panel.

A video demonstration of the scale's hardware and operation is available [at this link](#).

Calibration Curves and Comparison:

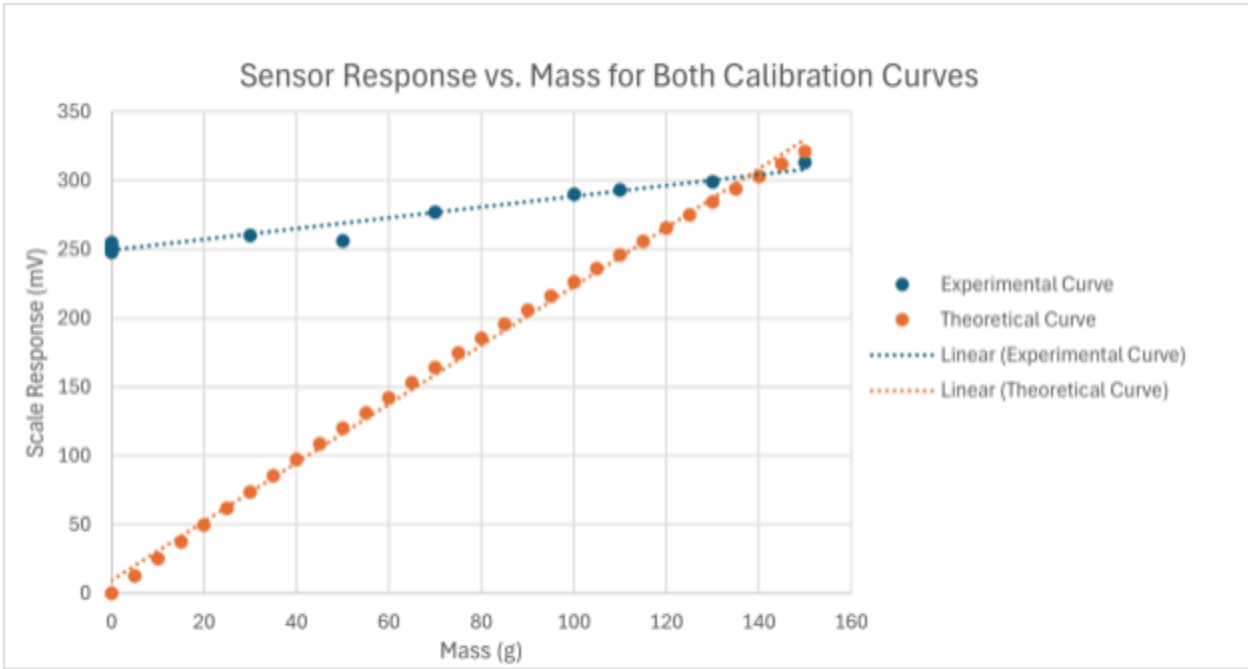


Figure 9: Original Theoretical Curve with Experimental Curve

Figure 9 above shows the original theoretical curve (pictured earlier in Figure 5) plotted with the experimental curve generated from measuring standards on the scale. The two curves are notably very different, which is a result of modifications made to the parameters of the scale (those listed in Table 1) to make construction feasible. Modifying the spring constant, distance, and resistance parameters to be more accurate to reality results in the graph pictured below in Figure 10. Specifically, the shunt resistor value was increased to account for the extra resistance coming from the excess nichrome length, the overall length of the scale was reduced match reality, the spring constant was increased to match the spring in the hardware, and the nichrome resistivity was reduced to match the nichrome used for construction. These updated parameters are pictured below in Table 2.

Table 2: Updated Parameter Values

Parameter	Value
g [gravity]	9.81 [m/s^2]
p [resistivity]	0.021 [Ω/m]
d1 [weight distance]	0.6 [m]
d2 [spring distance]	0.1 [m]
dc [CoM distance]	0.2 [m]
R1 [known resistor]	200 [Ω]

k [spring constant]	35 [N/m]
V0 [input voltage]	2.5 [V]
m _{self} [mass of scale bars]	0.1 [kg]

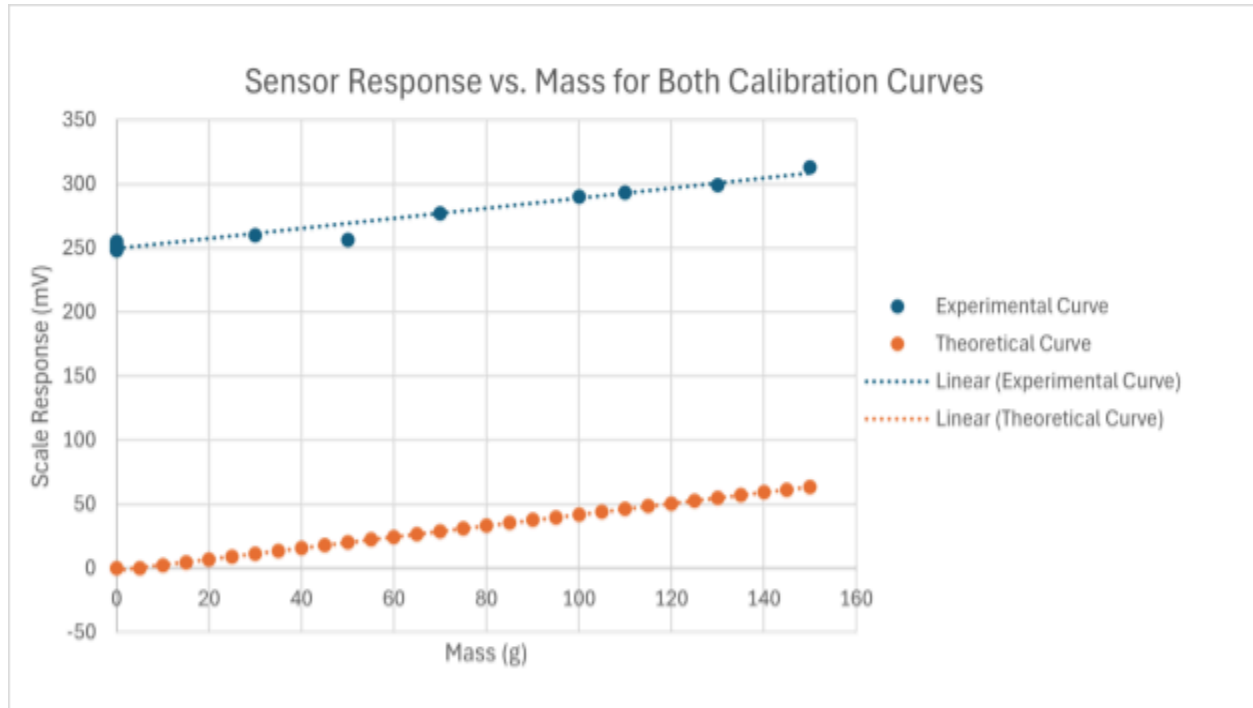


Figure 10: Updated Theoretical Curve with Experimental Curve

The updated theoretical curve appears to match the experimental curve fairly well except for the difference in y-intercept. This difference can be attributed to the resistance present across the DAQ leads when the scale is not affected by any mass, a consequence of having excess nichrome in series with the scale contact.

Determination of Static Sensitivity and Resolution:

In general, static sensitivity is defined as the slope of the calibration curve which is

$$S = \Delta Output / \Delta Input$$

However, because we used a linear line of best fit for our experimental calibration curve, we were able to determine the static sensitivity of our scale by using the m value of the calibration equation. This is also simply identified as the slope of the line, and was determined to be 0.391 mV/g.

The resolution of the scale is defined as the smallest change in input that creates a noticeable change in the output value. In order to find the actual value for our resolution we needed to utilize the error bars (uncertainty) for each point. The uncertainty was calculated by multiplying the standard deviation of the zero measurement values by the Student T value corresponding to our degrees of freedom and T-precision.

The standard deviation was calculated to be 0.002544056254 using an Excel spreadsheet (all data provided for calculations visible in appendix Figure a-3), and the t95 value corresponding to 5 degrees of freedom was found to be 2.571. Multiplying these values and then converting to volts (*1000mv/1V) gave us an error bar value of approximately 6.541.

After adding this error bar value to the calibration curve we found where the upper limit of the lowest bar on the zero measurement values intersected with the calibration curve and the corresponding x value in grams was found to be 25. This means the resolution of our scale design is 25 grams.

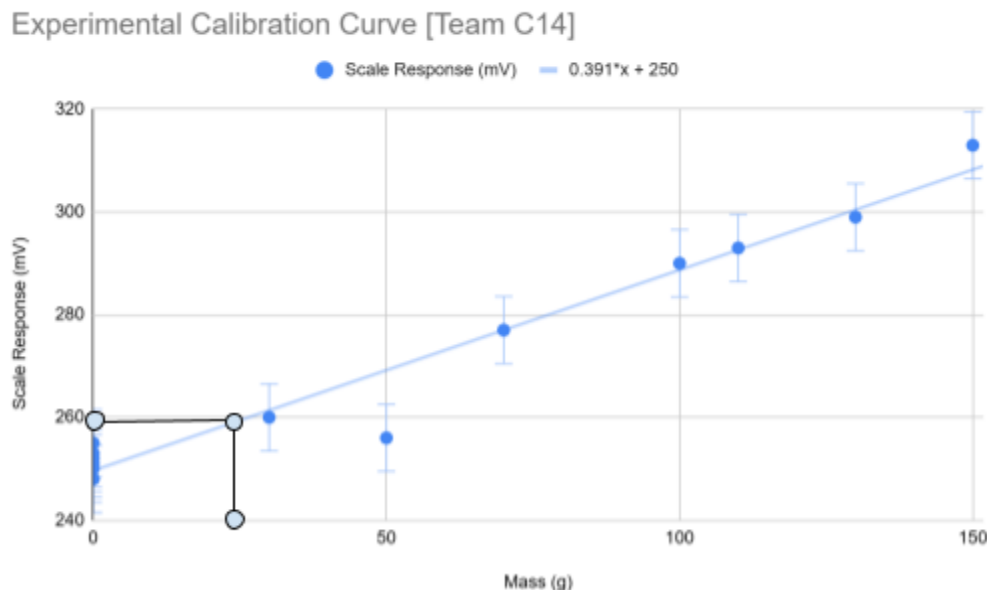


Figure 11: Experimental Calibration Curve with Error Bars and Resolution Indicated

Appendix:

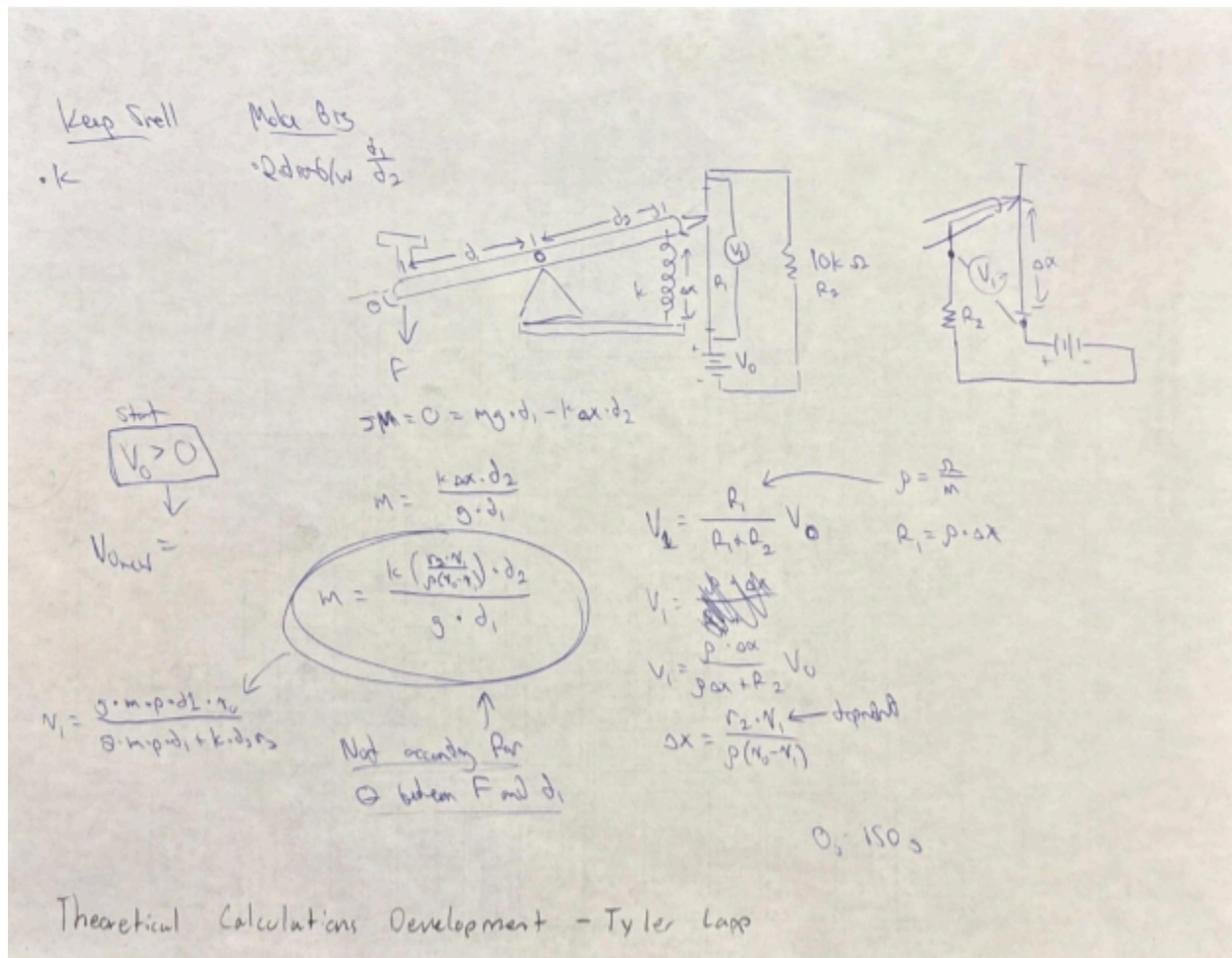


Figure a-1: Theoretical Calculations Development

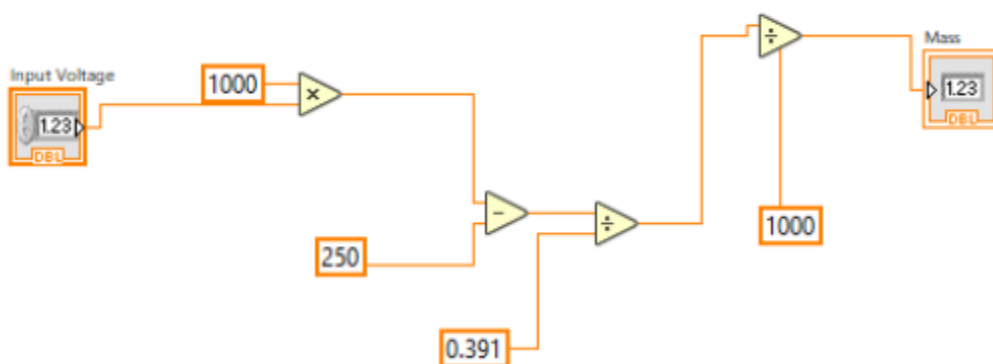


Figure a-2: Experimental Calibration Conversion Sub VI

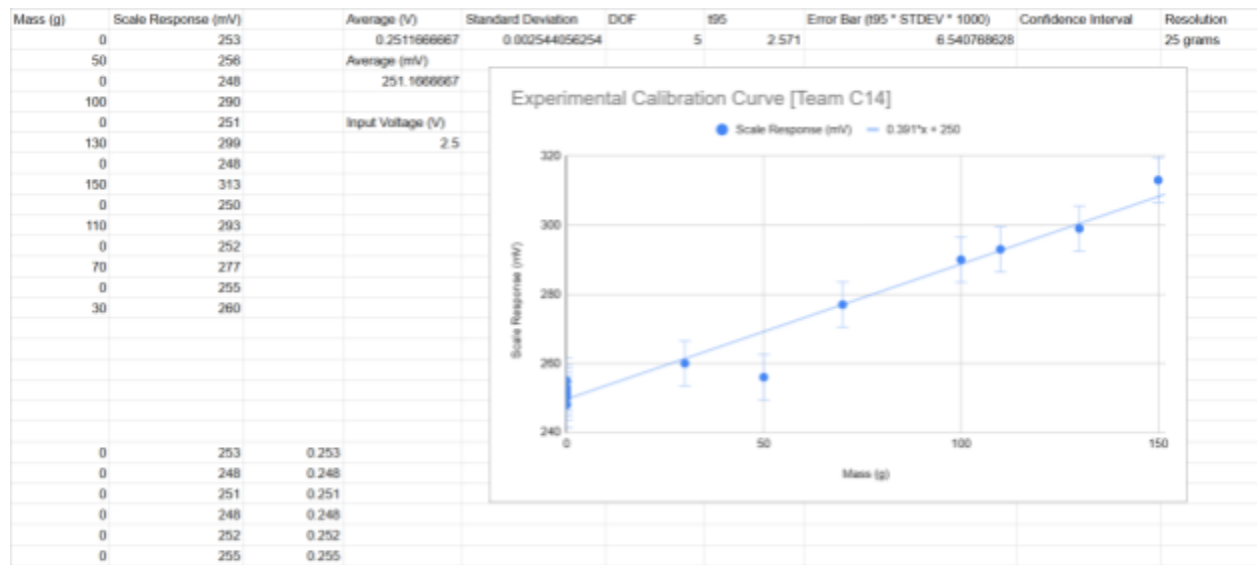


Figure a-3: Error Bar Development using Experimental Data